

MacroWatch

External Systems & References

Detailed guide to every external tool, API, and service used in MacroWatch — why selected, role, conditions of use

1. Anthropic Claude API

Why Selected

Claude provides the intelligence layer of MacroWatch — transforming raw data into coherent macro narratives, regime classifications, and investment positioning signals. Selected over GPT-4 for its stronger analytical reasoning on financial topics and more reliable JSON output format.

Role in MacroWatch

- Intel Tab: Generates full macro briefing (thesis, regime, alerts, key risks, watch list, Dalio lens, positioning) anchored to live market data
- Weekly Tab: Generates Fed policy narrative (Powell status, QT description) — numerical values are hardcoded separately
- QTR Tab: Generates government report summaries anchored to verified values
- Monthly Tab: Generates basis trade narrative

Technical Implementation

API Proxy: `api/claude.js` (Vercel serverless function) — proxies all Claude calls server-side to protect the API key. Adds system prompt enforcement that strips markdown and forces raw JSON output.

Client Helper: `src/api.js` — `callClaude(prompt, system)` function. Sets model, `max_tokens:4000`, parses JSON response. Falls back to `{raw: text}` if JSON parsing fails.

Model: `claude-haiku-4-5-20251001` — fastest and most cost-effective for structured JSON generation

Key: `VITE_ANTHROPIC_API_KEY` stored in Vercel environment variables

Cost & Limits

Item	Details
Pricing	Per token. Haiku is cheapest Claude model. Approximately \$0.25/million input tokens, \$1.25/million output tokens.
Max tokens per call	4,000 (set in <code>api.js</code>). Intel briefing typically 3,000-4,000 tokens.
Rate limits	Depends on plan. Free tier has strict limits. Paid tier recommended for production use.
Key management	Never expose in frontend code. Always proxy through <code>/api/claude.js</code> server-side.

2. GitHub

Why Selected

GitHub provides version control, code history, and the deployment trigger for Vercel. Every push to the main branch automatically triggers a Vercel deployment. GitHub also enables the iMac Terminal workflow for making code changes.

Role in MacroWatch

- Source code repository for all MacroWatch files
- Version control — every change is tracked with commit message and SHA
- Deployment trigger — Vercel watches the main branch and auto-deploys on every push
- Stable version tags — v1.0-stable, v1.1-stable allow rollback to known good states
- Documentation storage — /docs folder contains all PDF and markdown documentation

GitHub Personal Access Token — How It Works

The iMac Terminal uses GitHub's REST API to make code changes without needing git installed or SSH keys. A Personal Access Token (PAT) authenticates the API calls.

Step	Action
1. Generate token	github.com > Settings > Developer settings > Personal access tokens > Tokens (classic) > Generate new token
2. Export in Terminal	export GH_TOKEN='your_token_here' — must do this each new Terminal session as tokens are not persistent
3. Use in scripts	Python scripts read os.environ.get('GH_TOKEN','') and pass as Authorization header to GitHub API calls.
4. Make changes	Scripts: (a) fetch current file via GET, (b) modify content, (c) PUT updated content with sha of current file.
5. Vercel auto-deploys	Each successful GitHub push triggers Vercel build (15-20 seconds typical).
6. Token expiry	Tokens expire based on settings. Re-generate at github.com when expired. 401 Unauthorized = expired token

Repository Structure

Repository: <https://github.com/mlibrizzi/MacroWatch>

Owner: mlibrizzi | Branch: main | Vercel account: mlibrizzi-1604

3. Vercel

Why Selected

Vercel provides zero-configuration deployment for React/Vite apps with built-in serverless functions. The free hobby tier is sufficient for MacroWatch's usage. Vercel's edge network ensures fast load times globally, and automatic HTTPS makes PWA installation possible.

Role in MacroWatch

- Hosts the static React/Vite bundle (the frontend app)
- Runs serverless functions (/api/*.js) that proxy all external API calls
- Stores API keys securely as environment variables (never exposed in frontend)
- CDN edge caching for fast global delivery
- Automatic HTTPS — required for PWA installation on iPhone
- Auto-deploy from GitHub main branch — no manual deployment needed

Key Configuration

Setting	Value	Notes
Plan	Hobby (free)	Sufficient for personal use. No SLA.
Build command	npm run build (vite build)	Auto-detected from package.json
Output directory	dist	Vite default output
Node.js version	24.x	Set in deployment settings
Function region	iad1 (US East)	Closest to Treasury API servers
Cache purge	CDN > Caches > All content	Manual purge when stale data observed
Environment variables	Settings > Environment Variables	Keys: Anthropic, FRED, Finnhub, Gold API, OilPrice API

4. External Data APIs

API	URL	Data Provided	Plan / Cost	Rate Limits	Key Required	Notes
FRED	api.stlouisfed.org	Treasury yields, CPI, PCE GDP, VIX (2000-), Fed balance sheet, FOMC minutes, etc.	Free	1000 requests per minute	API Key	Historical data, real-time updates.
Finnhub	finnhub.io	Real-time stock quotes (Major US indices, 6000+ stocks), crypto prices, etc.	Free (Basic)	1000 requests per minute	API Key	Real-time data, historical data available.
gold-api.com	gold-api.com	Real-time gold (XAU/USD) and silver (XAG/USD) spot prices.	Free	1000 requests per minute	API Key	Spot price API. Real-time updates.
OilPriceAPI	api.oilpriceapi.com	Near real-time WTI crude oil (WTI_USD) and Brent oil (BRENT_USD) prices.	Free	1000 requests per minute	API Key	Oil prices, 24-hour updates.
open.er-api.com	open.er-api.com	EUR/USD, USD/JPY, GBP/USD exchange rates.	Free	1000 requests per minute	API Key	Simple FX API. Hourly updates on free plan.
Treasury Fiscal Data	treasdata.treasury.gov	Treasury auction results, Treasury securities (T-bills, T-notes, T-bonds).	Free	1000 requests per minute	API Key	Official US Treasury data. Real-time updates.
Treasury TIC	ticdata.treasury.gov	Major foreign holders of US Treasury securities.	Free	1000 requests per minute	API Key	Monthly data, 15-day lag. Releases ~45 days after month-end.
Anthropic API	api.anthropic.com	AI-generated macro narratives, regime classification, position, etc.	Paid (per token)	1000 requests per minute	API Key	Anthropic for AI. Full access requires paid plan.

5. APIs Evaluated But Not Used

API	Why Evaluated	Why Not Selected
Twelve Data	Tested for DXY and VIX. Symbols not available	DXY not available on free tier. Only ETF proxies returned. VIX not available
Alpha Vantage	Considered for DXY index data	Only 25 requests/day on free tier — insufficient for production use.
API Ninjas	Tested for WTI/Brent crude	Free tier only allows 7 rotating commodities per week — unreliable for production
EODHD	Evaluated for real-time stocks, DXY, and indices. Alpha (as EODHD) data (same as existing).	DXY available. Plan to integrate Alpha (as EODHD) data (same as existing).
OilPrice.com (direct)	Scraped for real-time oil	Not an official API — fragile.
Yahoo Finance (yfinance)	DXY.NYB ticker for DXY	No official API — scraping only. Unstable for production.
Bloomberg Terminal	True when-issued yield for auction	\$25,000/year — not viable.
Interactive Brokers API	True auction tail, futures curves, DXY	Account pending approval. Will integrate in Phase 3 when approved.

6. Development Tools

Tool	Version	Role	Notes
React	18.x	Frontend UI framework	Component-based. All UI in App.jsx.
Vite	5.x	Build tool and dev server	Fast HMR, optimized production builds.
Recharts	Latest	Chart library	Bar charts for auction demand trends.
Python 3	3.9 (macOS)	Deployment scripts	All GitHub API calls made via Python heredoc scripts in Terminal.
Node.js	24.x	Runtime for Vercel functions	Set in Vercel deployment settings.
reportlab	Latest	PDF generation	Used for creating this documentation.

7. Known Data Gaps & Future APIs

Gap	Impact	Planned Solution	Cost
Headline DXY (~98)	Showing Fed broad index (EOD) instead of DXY (Phase 2)	EODHDX (Phase 2)	~\$22/month
True auction TAIL	Showing N/A — most important auction Bids were issued ytd (Phase 2)	Interactive Bids were issued ytd (Phase 2)	~\$0 (Phase 2)
Futures curves	No gold/Treasury/oil forward pricing	Interactive Brokers API (Phase 3)	Included with IB account
5Y5Y breakeven	Inflation expectations not monitored	FRED series T5YIFR (free, Phase 2)	\$0
Corporate credit spreads	Stress indicator missing	FRED series BAMLH0A0HYM2 (free, Phase 2)	\$0
Bitcoin price	Dalio emerging hard money crypto	QuinGekissingFinnhub (free, Phase 2)	\$0
Real-time EUR/USD	Hourly delay on FX	EODHD forex (Phase 2)	~\$22/month
Historical sparklines	No trend visualization	FRED historical + EODHD (Phase 2)	~\$22/month